GT
05

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[redacted]

Instructions:

Date: 24-04-26

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on side-1 and side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The CODE for this Booklet is E1. Make sure that the CODE printed on Side-2 of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is NOT permissible on the Answer Sheet.
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

1. The suffix used for families in ICZN and ICBN, respectively, are

(1) aceae, idea (2) phyta, ales

(3) ae, nae (4) idae, aceae

2. Select the barrier methods of contraception.

A. Condoms

B. Diaphragm

C. Vaults

D. Multiload–375

E. LNG – 20

(1) A, B, and D (2) A, B, and E

(3) A, B, and C (4) A and B only

3. Given below is a photosynthetic reaction.

$$H_2A + CO_2 \rightarrow CH_2O + H_2O + A$$

If this reaction occurs in green-sulphur bacteria, then what is A?

(1) S (2) H O

(3) N (4) O

4. Which of the following is ***not*** a protein?

(1) RuBisCO (2) Diterpenes

(3) Collagen (4) Trypsin

5. Two butterfly species are competing for the same nectar of a flower in a garden. To survive and co-exist together, they may avoid competition in the same garden by

(1) feeding at the same time

(2) choosing different foraging patterns

(3) increasing the time spent on attacking each other

(4) predating on each other

6. Which one of the following statements is ***incorrect***?

(1) In cockroaches and prawns, excretion of waste material occurs through Malpighian tubules.

(2) In ctenophores, locomotion is brought about by comb plates.

(3) In *Fasciola*, flame cells help in excretion.

(4) Leeches are hermaphrodites and exhibit internal fertilisation.

7. Which of the following statements are correct about vital capacity?

A. It includes ERV, TV, and IRV.

B. It is the total volume of air a person can inspire after a normal expiration.

C. It is the maximum volume of air a person can breathe in after forced expiration.

D. It includes ERV, RV, and IRV.

E. It is the maximum volume of air a person can breathe out after a forced inspiration.

(1) B, D, and E (2) A, C, and D

(3) A, C, and E (4) A and E

8. Given below are two statements.

Statement I : The organisation of prokaryotic cells is fundamentally similar, even though prokaryotes exhibit a wide variety of shapes and functions.

Statement II : The shape of cells may vary with the function they perform.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

9. Given below are two statements.

Statement I : In frogs, a neck is present, but a tail is absent.

Statement II : In frogs, eyes are bulged and covered by a nictitating membrane that protects them while in water.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

10. If living cells, similar to those found on earth, were found on another planet, where there is no oxygen, then which cell organelle would most probably be absent?

(1) Ribosomes

(2) Golgi apparatus

(3) Mitochondria

(4) Endoplasmic reticulum

11. A female individual shows a small, round head, furrowed tongue, partially open mouth, and broad palms with a characteristic palmar crease. She also exhibits delayed physical, psychomotor, and mental development. The karyotype analysis of such an individual is most likely to reveal

(1) 47 chromosomes with XXY sex-chromosomes

(2) 45 chromosomes with XO sex – chromosomes

(3) 47 chromosomes with XYY sex – chromosomes

(4) Trisomy of chromosome 21

12. Given below are two statements.

Statement I : Both Bladderwort and *Cuscuta* are insectivorous plants.

Statement II : In *Agaricus*, basidiospores are produced endogenously in basidium.

Choose the correct option from the following.

(1) Statement I is correct, but statement II is incorrect.

(2) Statement I is incorrect, but statement II is correct.

(3) Both statements I and II are correct.

(4) Both statements I and II are incorrect.

13. Which of the following are correctly matched?

A. Cocaine – Interferes with the transport of the neurotransmitter dopamine.

B. Benzodiazepines – Used in the treatment of insomnia.

C. Tobacco smoke – Lung cancer

D. Marijuana – Generally taken by injection

(1) A, B, and C only

(2) B, C, and D only

(3) A, C, and D only

(4) A and C only

14. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Many nations find *in situ* conservation unrealistic and economically not feasible to conserve all their biological wealth.

Reason (R) : Invariably, the number of species waiting to be saved from extinction far exceeds the conservation resources available.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

15. Arrange the events that occur in the life cycle of the gymnosperms in appropriate order.

A. Release of male gametophyte from microsporangium.

B. Fusion of male and female gamete leads to the formation of a zygote.

C. Development of the embryo inside the multicellular female gametophyte within the megasporangium.

D. The pollen tube discharges its contents in the mouth of the archegonia.

E. Male gametes are carried through the pollen tube towards the archegonia in the ovules

Choose the correct answer from the options given below.

(1) A → E → D → B → C

(2) C → A → E → B → D

(3) C → E → A → B → D

(4) E → A → C → D → B

16. Read the following statements.

A. The phylogenetic system of classification assumes that the organisms belonging to the same taxa have a common ancestor.

B. The complete flowers are not necessarily bisexual.

C. Sex organs in bryophytes are multicellular and jacketed.

D. Spores give rise to inconspicuous, free-living, small but multicellular prothallus in pteridophytes.

E. Pollination is a prerequisite for fertilisation in seed-bearing plants.

How many of the above statements are correct?

(1) Two (2) Four

(3) One (4) Five

17. Match List I with List II.

	List I		**List II**
A.	Insects	I.	According to the IUCN Red List, this group appears to be more vulnerable to extinction
B.	Reptiles	II.	This group makes up more than 70% of the total animal species.
C.	Birds	III.	378 species of this group are found in the tropical Amazon rainforest.
D.	Amphibians	IV.	Nearly 1400 species of this group are present in Colombia.

Choose the correct answer from the options given below.

(1) A-III B-II C-I D-IV

(2) A-IV B-II C-III D-I

(3) A-II B-III C-IV D-I

(4) A-II B-IV C-III D-I

18. The following are the floral features. Bracteate, actinomorphic, unisexual flowers with five united sepals, bell shaped and five united petals showing valvate aestivation, and a pentacarpellary, apocarpous gynoecium with an inferior ovary. Select the correctly represented floral formula.

(1) $\text{Ebr} \oplus \female\ \text{K}_{(5)}\,\text{C}_5\,\text{A}_o\,\overline{\text{G}}_5$

(2) $\text{Ebr} \oplus \female\ \text{K}_5\,\text{C}_{(5)}\,\text{A}_o\,\overline{\text{G}}_5$

(3) $\text{Br}\ \% \male\ \text{K}_5\,\text{C}_5\,\text{A}_5\,\underline{\text{G}}_5$

(4) $\text{Br} \oplus \female\ \text{K}_{(5)}\,\text{C}_{(5)}\,\text{A}_o\,\overline{\text{G}}_5$

19. Which of the following statements about the midbrain of humans is ***not*** correct?

(1) It is located between the thalamus/ hypothalamus and pons.

(2) It contains arbor vitae.

(3) It has a canal called the cerebral aqueduct.

(4) Its dorsal part consists of 4 optic lobes (corpora quadrigemina).

20. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Simple leaves and compound leaves can be distinguished from each other on the basis of the presence or absence of axillary buds.

Reason (R) : Axillary buds are present in the axil of both simple and compound leaves, but absent in the axil of leaflets of compound leaves.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

21. In a healthy pregnant woman in her first trimester, laboratory assessment of total gonadotropin activity is carried out. Which of the following findings is expected?

(1) A very high concentration of FSH and LH circulating in the uterus to promote implantation of the embryo.

(2) A very high concentration of circulating hCG that maintains the corpus luteum, thereby ensuring continued secretion of oestrogen and progesterone

(3) A very high concentration of circulating hCG, whose primary role is to thicken the myometrium.

(4) Very high levels of FSH and LH within the uterus, whose main function is to thicken the endometrium.

22. Read the following statements and find out the ***incorrect*** set of statements.

A. Root hairs originate from the epiblema, while lateral roots originate from the endodermis.

B. During secondary growth in a dicot root, the cambium ring develops from the pericycle outer to xylem strands and from conjuctive tissue in between the xylem and phloem strands.

C. In dicot roots, suberin is deposited in the tangential and radial walls of barrel-shaped endodermal cells.

D. Pith is distinct and well-developed in the sunflower root and maize stem.

E. The outer walls of guard cells are thick, and the inner walls (towards pore) are much thinner.

(1) A, C, and D only

(2) A, D, and E only

(3) A, B, D, and E only

(4) A and E only

23. Which of the following statements is ***incorrect*** regarding a standard electrocardiogram?

(1) QRS complex indicates ventricular depolarisation.

(2) P-wave indicates depolarisation of atria.

(3) Time between S and T represents atrial systole.

(4) End of T-wave indicates the end of ventricular contraction.

24. Match List I with List II.

	List I		**List II**
A.	Promoter of catalytic activity of enzyme	I.	Amino acid
B.	Possess glycosidic bond	II.	Cofactor
C.	Storage polysaccharide in animal	III.	Glycogen
D.	Primary metabolite	IV.	Starch

Choose the correct answer from the options given below.

(1) A-II B-IV C-III D-I

(2) A-III B-I C-IV D-II

(3) A-III B-IV C-I D-II

(4) A-II B-III C-I D-IV

25. Given below are two statements.

Statement I : Formation of neoplastic cells from normal cells may be induced by certain agents called carcinogens.

Statement II : Antibodies against cancer-specific antigens are also used for the detection of certain cancers.

Choose the correct option from the following.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

26. Choose the correct set of bones of the appendicular skeleton in humans.

(1) Humerus, scapula, sternum, collar bone

(2) Clavicle, coxal bone, sphenoid, coccyx.

(3) Sacrum, zygomatic, radius, carpals

(4) Carpals, patella, fibula, scapula

27. Read the following statements regarding the nucleolus and choose the correct option.

A. Is a non-membrane-bound structure

B. Is larger and more numerous in cells undergoing protein synthesis

C. Is the site of 70S ribosome assembly

(1) Only (A) is true.

(2) Only (B) is true.

(3) Both (A) and (B) are true.

(4) (A), (B), and (C) are true.

28. Sexual deceit in *Ophrys* is associated with which of the following?

A. Mimicry, where a male wasp resembles a flower

B. Uncanny resemblance of a petal of the flower to the female bee

C. Pseudocopulation

D. Entomophily

Choose the correct answer from the options given below.

(1) Only A and D

(2) Only B and C

(3) All except A

(4) Only C and D

29. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : A thermostable DNA polymerase is used in PCR for repeated amplification.

Reason (R) : PCR involves repeated cycles of high temperature for DNA denaturation, and thermostable DNA polymerase can withstand high temperatures without denaturation.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

30. Select the ***incorrect*** match.

(1) Ascariasis – Internal bleeding

(2) Filariasis – Chronic inflammation of the organs

(3) Ringworms – Scaly lesions on various parts of the body.

(4) Amoebiasis – Chill and high fever recurring every 3 to 4 days.

31. If the amount of DNA in a sperm cell is 20 Pg, then find out the amount of DNA in the meiocyte of the same organism in G2 phase, anaphase I and each cell of anaphase II (in Pg)?

(1) 80, 80, 40

(2) 40, 40, 20

(3) 20, 40, 40

(4) 80, 40, 20

32. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : The human heart is myogenic.

Reason (R) : In the human heart, adjacent cardiac muscle fibres communicate with each other through intercalated discs.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

33. Read the following statements.

A. The formation of a new cell wall begins with the formation of a simple precursor called the cell plate that represents the middle lamellum in plant cells.

B. Failure of karyokinesis after cytokinesis results in the formation of a syncytium.

C. Meiosis occurs in diploid cells.

D. Apical meristem undergoes mitotic division while vascular cambium undergoes meiotic division.

E. Mitotic cell division occurs to restore the nucleocytoplasmic ratio of a cell, which gets disturbed during cell growth.

How many statements are ***incorrect***?

(1) 4 (2) 2

(3) 3 (4) 5

34. Match List I with List II.

	List I		List II
A.	Rapid spasm	I.	Inflammation of joints
B.	Myasthenia gravis	II.	Weakening and paralysis of skeletal muscles
C.	Arthritis	III.	Genetic disorder
D.	Muscular dystrophy	IV.	Wild contraction

Choose the correct answer from the options given below.

(1) A-I B-II C-III D-IV

(2) A-IV B-III C-II D-I

(3) A-III B-I C-IV D-II

(4) A-IV B-II C-I D-III

35. Select the ***incorrect*** combination of an animal, its heart and the type of circulation.

(1)	*Rohu*	Two-chambered heart	Single circulation
(2)	Frog	Three-chambered heart	Incomplete double circulation
(3)	Lizard	Three-chambered heart	Single circulation
(4)	Human	Four-chambered heart	Double circulation

36. Xylem parenchyma stores food material in the form of

(1) starch and resins

(2) fat and mucilage

(3) starch and fat

(4) tannins and latex

37. Given below are two statements.

Statement I : Fertilisation can only occur if the ovum and sperm are transported simultaneously to the birth canal.

Statement II : In ectopic pregnancy, the blastocyst can implant in the fallopian tubes.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

38. Tegmina in cockroaches arise from

(1) mesothorax

(2) metathorax

(3) prothorax and mesothorax

(4) prothorax

39. Given below is the sequence of amino acids from N-terminus to C-terminus coded by mRNA in an organism.

Histidine – Cysteine – Methionine – Tryptophan – Aspartic acid

Codons for the given amino acids are as follows.

His – CAU or CAC, Cys – UGU or UGC, Met – AUG, Trp – UGG, Asp – GAU or GAC

Which among the following can be a sequence of the template in DNA that codes for mRNA for the given peptide?

(1) 3'- GTCCCACATGCAGTG -5'

(2) 5'- GTCCCATACGCAGTG - 3'

(3) 5'- GTCCCACATGCAGTG - 3'

(4) 3' -GTCCATCCGGCAGTG – 5'

40. The enzyme named carbonic anhydrase causes the formation of carbonic acid from CO_2 and water. Starting from 10°C, its activity doubles for every 10°C rise till 40°C, after which it falls rapidly. Which of the following types of curves is obtained when enzyme activity is plotted against temperature?

(1) Bell-shaped

(2) Linear rise

(3) Rectangular

(4) Sigmoid

41. Match List I with List II.

	List I		**List II**
A.	Gene therapy	I.	Amplification of nucleic acids
B.	PCR	II.	Detection of pathogen, based on antigen–antibody reaction
C.	ELISA	III.	Treatment for ADA deficiency
D.	DNA probe	IV.	Detection of complementary DNA sequence by hybridisation

Choose the correct answer from the options given below.

(1) A-III B-II C-I D-IV

(2) A-IV B-II C-III D-I

(3) A-III B-I C-II D-IV

(4) A-IV B-III C-II D-I

42. Post-transcriptional modifications in eukaryotes include all, except

(1) addition of 7 methyl guanosine triphosphate cap

(2) addition of poly A sequence

(3) splicing of introns

(4) removal of UTRs from mRNA

43. Choose the ***incorrectly*** matched pair from the following.

(1) Areolar connective tissue – Most widely distributed connective tissue in our body.

(2) Adipose connective tissue – Good insulator and thus can reduce heat loss through the skin.

(3) Dense irregular connective tissue – Occurs in the dermis of the skin.

(4) Fluid connective tissue – Contains cells that secrete modified polysaccharides and collagen fibres.

44. A plasmid named pBR322 possesses *ampR* and *tetR* genes along with the restriction sites for certain restriction enzymes. If a foreign gene of interest is inserted at a site 'X', and the recombinant plasmid is transferred into *E. coli* and plated on a suitable media, then the following results are obtained.

Medium	**Growth of recombinant**
Ampicillin	+
Tetracycline	–

The above results indicate that the foreign gene of interest is most likely ligated with the restriction enzyme named

(1) Cla I (2) Sal l

(3) Pvul (4) EcoRI

45. Study the following.

A. The proximity between the Henle's loop and the vasa recta.

B. The flow of filtrate in the two limbs of Henle's loop is in opposite directions.

C. The flow of blood through the two limbs of vasa recta follows a countercurrent pattern.

D. The selective secretion of bicarbonate and hydrogen ions in PCT.

Which of the above factors help in maintaining an increasing osmolarity towards the inner medullary interstitium of the human kidney?

(1) A, B, and C only (2) B and D only

(3) B, C, and D only (4) A and B only

46. Pomato, a new variety plant is developed through

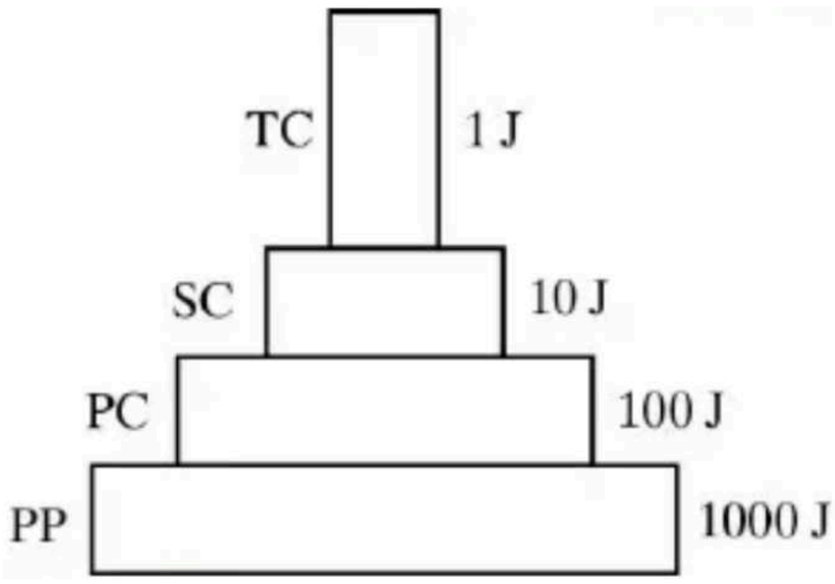
(1) callus culture

(2) somatic embryogenesis

(3) somatic hybridisation

(4) meristem culture

47. Study the following diagram.



The above is an ideal pyramid of energy of an ecosystem. What is the total amount of sunlight coming into the ecosystem?

(1) 1,000,000 J (2) 1,00,000 J

(3) 10,000 J (4) 100 J

48. For isolation of DNA from the plant cell, which of the following enzyme is ***not*** used?

(1) Deoxyribonuclease

(2) Cellulase

(3) Ribonuclease

(4) Protease

49. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Animals of phylum Echinodermata are commonly known as spiny-bodied animals.

Reason (R) : These animals have an exoskeleton made of calcareous ossicles.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

50. Cucumber is treated with a hormone that promotes the female flowers, thus increasing the yield. Identify that phytohormone.

(1) GA3 (2) Ethylene

(3) IAA (4) Zeatin

51. Which of the following disorders are considered Mendelian disorders?

A. Cystic fibrosis

B. Klinefelter's syndrome

C. Sickle cell anaemia

D. Thalassemia

E. Myotonic dystrophy

Choose the correct answer from the options given below.

(1) A, B, and E only

(2) C and D only

(3) A, C, D, and E only

(4) B and E only

52. Given below are two statements.

Statement I : A stirred-tank bioreactor is usually cylindrical or has a curved base to facilitate the mixing of the reactor contents.

Statement II : The downstream processing and quality control testing vary from product to product.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

53. Passive immunity is slow, takes weeks to develop and provides long-lasting protection, as seen in vaccination. Similarly, active immunity provides fast protection by directly administering preformed antibodies (or) antitoxins, but it is short-lived.

Choose the correct answer from the following.

(1) The statement is true for both active and passive immunity.

(2) The statement is true for active immunity but false for passive immunity.

(3) The statement is false for both active and passive immunity.

(4) The statement is false for active immunity but true for passive immunity.

54. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Each cell of sporogenous tissue in a microsporangium is a potential pollen or microspore mother cell.

Reason (R) : Each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

55. Which of the following statements is correct?

(1) Nearly 70% of carbon dioxide is transported in a physically dissolved form in plasma.

(2) Carbonic anhydrase occurs in very high concentration in RBCs but only in minute quantities in plasma.

(3) Formation of carbamino haemoglobin is primarily regulated by the partial pressure of oxygen.

(4) About 8 mL of CO_2 is delivered to the alveoli by every 100 mL of deoxygenated blood under normal physiological conditions.

56. Arrange the following terms in the correct developmental sequence in angiosperms

A. Pollen grain

B. Sporogenous tissue

C. Microspore tetrad

D. Pollen mother cell

E. Male gametes

Choose the correct answer.

(1) (A) → (B) → (C) → (D) → (E)

(2) (C) → (E) → (A) → (D) → (B)

(3) (B) → (D) → (C) → (A) → (E)

(4) (B) → (C) → (A) → (D) → (E)

57. If a colourblind female marries a man whose mother was also colourblind, what are the chances of her progeny having colour blindness?

(1) 50% (2) 75%

(3) 100% (4) 25%

58. Match List I with List II.

List I		List II	
A.	Monoadelphous	I.	*Citrus*
B.	Diadelphous	II.	Pea
C.	Polyadelphous	III.	Lily
D.	Epiphyllous	IV.	China-rose

Choose the correct answer from the options given below.

(1) A-IV B-I C-II D-III

(2) A-IV B-II C-I D-III

(3) A-I B-II C-IV D-III

(4) A-III B-I C-IV D-II

59. Match List I with List II.

List I		List II	
A.	Primary follicle	I.	Antrum
B.	Tertiary follicle	II.	Ovulation
C.	Graffian follicle	III.	Mitochondria in the middle piece of sperm
D.	Nebenkern	IV.	Granulosa cells

Choose the correct answer from the options given below.

(1) A-IV B-II C-I D-III

(2) A-I B-IV C-III D-II

(3) A-IV B-I C-II D-III

(4) A-II B-I C-IV D-III

60. Which of the following blood group/s is/are possible when both parents possess the AB blood group?

(1) A, B, and AB

(2) A, B, and O

(3) B only

(4) A, B, AB, and O

61. Identify the ***incorrect*** statement.

(1) Abortions could happen spontaneously too.

(2) Infertility is defined as the inability to produce a viable offspring and is always due to defects in the female partner.

(3) Complete lactation up to 6 months following parturition could help as a natural method of contraception.

(4) Creating awareness about sex-related aspects is an effective method to improve the reproductive health of people.

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62. Select the odd one out w.r.t. recessive trait in pea plants.

(1) Green seed colour

(2) Green pod colour

(3) Yellow pod colour

(4) White flower colour

63. A couple unable to conceive has normal gamete production, but the female has severely damaged fallopian tubes, preventing normal embryo transport. Which of the following ART techniques would be most appropriate for their situation?

(1) GIFT, as it bypasses damaged tubes completely.

(2) ZIFT, as early embryos bypass the damaged region.

(3) Artificial insemination, as it deposits sperm in the uterus.

(4) IVF followed by IUT, as the embryo is placed directly in the uterus.

64. Read the following statements.

A. Flowers are morphological and embryological marvels and the site of sexual reproduction.

B. Flowering is initiated by several hormonal and structural changes.

C. The number and length of stamens are variable in flowers of different species.

D. Four lobes are present in one anther.

How many of the above statements are correct?

(1) One

(2) Three

(3) Two

(4) Four

65. Given below are two statements.

Statement I : A population growing in a habitat with limited resources initially shows a lag phase, followed by phases of acceleration and deceleration and finally an asymptote.

Statement II : Every population in nature has at its disposal unlimited resources to permit exponential growth.

In light of the above statements, choose the correct answer from the options given below.

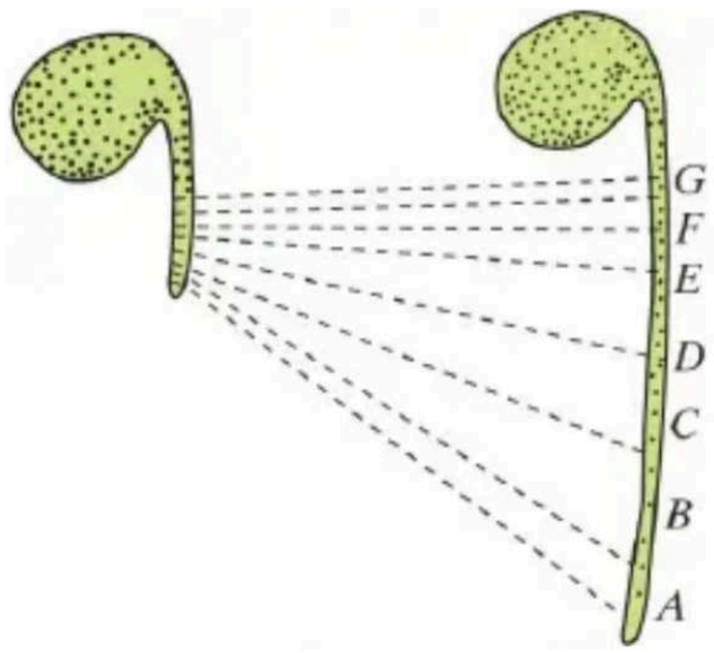
(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

66. The diagram given below represents



(1) parallel line technique to detect stem thickness

(2) effect of auxin on root growth

(3) radial line technique to detect the meristematic zone

(4) parallel line technique to detect the zone of elongation.

67. Match List I with List II regarding the salient features of the human genome.

	List I		List II
A.	Human genome size	I.	Less than 2%
B.	Total number of genes	II.	3,164.7 million bp
C.	Percentage of genome coding for proteins	III.	99.9 %
D.	Nucleotide sequence similarity in humans	IV.	Approximately 30,000

Choose the correct answer from the options given below.

(1) A-II B-III C-I D-IV

(2) A-II B-IV C-I D-III

(3) A-I B-II C-IV D-III

(4) A-III B-I C-II D-IV

68. Read the following statements.

A. Two nucleotides are linked through 3'-5' phosphodiester linkage to form a dinucleotide.

B. Lambda phage has single-stranded linear DNA with 48502 nucleotides.

C. Rho factor (ρ) is required for termination of transcription.

D. UTRs are required for efficient transcriptional processes.

E. M. Wilkins and Rosalind Franklin had proposed a definitive double helix model of DNA on the basis of X-ray diffraction pictures in 1953.

Choose the correct option.

(1) Only A and C are correct.

(2) Only B, D, and E are correct.

(3) Only B and C are correct.

(4) A, B, C, D, and E are correct.

69. Select the ***incorrectly*** matched option among the following regarding disorders of the excretory system.

(1)	Renal calculi	Stone (or) insoluble mass of crystallised salts formed within the kidney
(2)	Glomerulo – nephritis	Inflammation of glomeruli of kidney
(3)	Glycosuria	Presence of glucose in blood
(4)	Ketonuria	Presence of ketone bodies in urine

70. Two biomolecules, *A* and *B*, were isolated from a cell extract. *A* yielded only a nitrogenous base and sugar upon mild acid hydrolysis.

B also released a nitrogenous base and sugar upon hydrolysis, and on further treatment with alkaline phosphatase, inorganic phosphate was also obtained.

Based on the information given, we can correctly conclude that

(1) *A* is a nucleoside and *B* is a nucleotide.

(2) *B* is a nucleoside and *A* is a nucleotide.

(3) *B* contains phosphodiester bonds that get cleaved by alkaline phosphatase to liberate phosphate.

(4) On treatment of *A* with mild acid, the ester bond present between the hexose sugar and nitrogenous base gets hydrolysed.

71. Given below are two statements.

Statement I : About 350 mya, fish with stout and strong fins could move on land and go back to water.

Statement II : In 1998, a fish caught in South America happened to be a coelacanth which was thought to be extinct.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

72. Read the following.

A. Enlargement of the thyroid gland

B. Protrusion of the eyeballs

C. Increased basal metabolic rate

D. Weight loss

All the above conditions are indicative of which disease?

(1) Hypothyroidism

(2) Cretinism

(3) Graves' disease

(4) Tetany

73. Match List I with List II.

	List I		**List II**
A.	Big Bang theory	I.	Formation of organic molecules from inorganic precursors
B.	Chemical evolution	II.	Origin of life from decaying matter
C.	Panspermia	III.	Formation of the universe by a single huge explosion
D.	Spontaneous generation	IV.	Origin of life from spores arriving from outer space

Choose the correct answer from the options given below.

(1) A-III B-I C-IV D-II

(2) A-III B-I C-II D-IV

(3) A-IV B-I C-II D-III

(4) A-IV B-III C-II D-I

74. The hormones that regulate gene expression by interaction of the hormone-receptor complex with genome

(1) interact with membrane-bound receptors

(2) act through second messenger cAMP

(3) are mostly steroids in their chemical nature

(4) are all derived from cholesterol only

75. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : All green plants fix CO_2 through C_3 pathway, but not all plants are C_3 plants.

Reason (R) : C_3 pathway or Calvin pathway is universal to all photosynthetic plants; however, whether a plant is C_3 or C_4 is determined by its first stable product of carbon dioxide fixation.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

76. Termination of translation requires

(1) sigma factor (σ)

(2) rho factor (ρ)

(3) operator

(4) release factor

77. Which of the following is chemically and functionally different from the others?

(1) Selectable marker

(2) Eco RI

(3) Bam HI

(4) Pst I

78. What is the average rate of polymerisation of DNA-dependent DNA polymerase in *E. coli*?

(1) 2000 bases/hour

(2) 2000 bps/second

(3) 2000 bps/minute

(4) 1000 bps/minute

79. Match List I with List II.

	List I		List II
A.	Beta-galactosidase	I.	Enzymatic synthesis of RNA
B.	Permease	II.	Peptide bond formation
C.	Polynucleotide phosphorylase	III.	Hydrolysis of lactose
D.	Ribozyme	IV.	Increases permeability of cell to Beta-galactosides

Choose the correct answer from the options given below.

(1) A-III B-IV C-I D-II

(2) A-IV B-III C-II D-I

(3) A-IV B-III C-I D-II

(4) A-II B-I C-III D-IV

80. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : The respiratory quotient depends upon the type of respiratory substrate used during respiration.

Reason (R) : When carbohydrates are used as substrate and are completely oxidised, the RQ will be 1, and in case of fats, it will be 0.7.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

81. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : A patient with ADA deficiency requires periodic infusion of genetically engineered lymphocytes that can produce functional ADA.

Reason (R) : The lymphocytes have a limited life span, and their death causes enzyme deficiency again in the patient.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

82. Consider the following statements.

A. The water vascular system of echinoderms helps in locomotion, whereas the canal system of sponges does not.

B. All eucoelomate animals exhibit bilateral symmetry during their adult stages.

C. All birds have feathers, whether they fly or not.

D. All diploblastic animals lack coelom.

Which of the above are correct?

(1) A and B

(2) A, C, and D

(3) B, C, and D

(4) B and D

83. Match List I with List II.

	List I		List II
A.	4C	I.	Acetyl CoA
B.	2C	II.	Pyruvate
C.	5C	III.	Citric acid
D.	3C	IV.	α –ketoglutaric acid
		V.	Malic acid

Choose the correct answer from the options given below.

(1) A-II B-V C-III D-I

(2) A-III B-I C-IV D-II

(3) A-V B-I C-IV D-II

(4) A-IV B-III C-I D-II

84. What is the range of 'Z' in species–area relationships when analysed over large areas such as an entire continent?

(1) 0.1 to 0.2

(2) 0.6 to 1.2

(3) 0.1 to 0.6

(4) 0.2 to 0.6

85. Gross primary productivity minus respiratory losses is equal to the

(1) rate of formation of new organic matter by consumers

(2) rate of production of organic matter during photosynthesis

(3) biomass available to herbivores and decomposers for consumption

(4) amount of biomass (or) organic matter produced per unit area over a time period by plants during photosynthesis

86. Which of the following are correct regarding mutations according to Hugo de Vries?

A. Mutations arise suddenly in a population.

B. Mutation causes evolution.

C. Mutations are random.

D. Mutations are large variations.

E. Mutations are directionless.

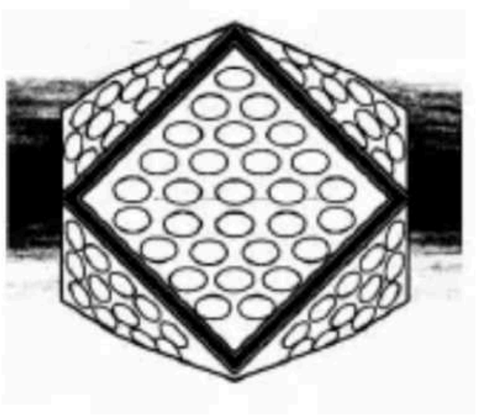
(1) A, B, and C only

(2) A, D, and E only

(3) B, D, and E only

(4) A, B, C, D, and E

87. Which of the following diseases is caused by the microbe shown in the diagram given below?



(1) Mosaic disease

(2) Foot and mouth disease

(3) Mad cow disease

(4) Respiratory infection

88. Given below are two statements.

Statement I : Respiratory substrates are often more than one in living organisms.

Statement II : Pure proteins or fats are used as respiratory substrates.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

89. Match List I with List II.

	List I		List II
A.	Cell in Watermelon	I.	Measured in terms of size of the cell
B.	Root apical meristem of maize grain	II.	Measured in terms of length
C.	Growth of a pollen tube	III.	Measured in terms of surface area
D.	Growth in dorsiventral leaf	IV.	Measured in terms of cell number

Choose the correct answer from the options given below.

(1)	A-I	B-IV	C-II	D-III
(2)	A-IV	B-II	C-I	D-III
(3)	A-II	B-III	C-IV	D-I
(4)	A-III	B-I	C-II	D-IV

90. Given below are two statements.

Statement I : Mendel's study found that the factors independently assort and combine in all permutations and combinations.

Statement II : Mendel's law of independent assortment does not hold true for the genes located on the same chromosomes.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

91. A motorcycle and a car start from rest from the same place at the same time and travel in the same direction. The motorcycle accelerates at 1.0 m/s^2 up to a speed of 36 km/h and the car at 0.5 m/s^2 up to a speed of 54 km/h. The time at which the car would overtake the motorcycle is

(1) 20 s (2) 25 s

(3) 30 s (4) 35 s

92. Dimensions of $\frac{1}{\mu_0 \varepsilon_0}$, where symbols have their usual meaning, are

(1) $[L^{-2}T^2]$ (2) $[L^2T^{-2}]$

(3) $[L^1T^{-1}]$ (4) $[L^{-1}T^1]$

93. The mass and volume of a body are 4.237 g and 2.5 cm^3, respectively. The density of the material of the body in correct significant figures is

(1) 1.6948 g cm^{-3} (2) 1.69 g cm^{-3}

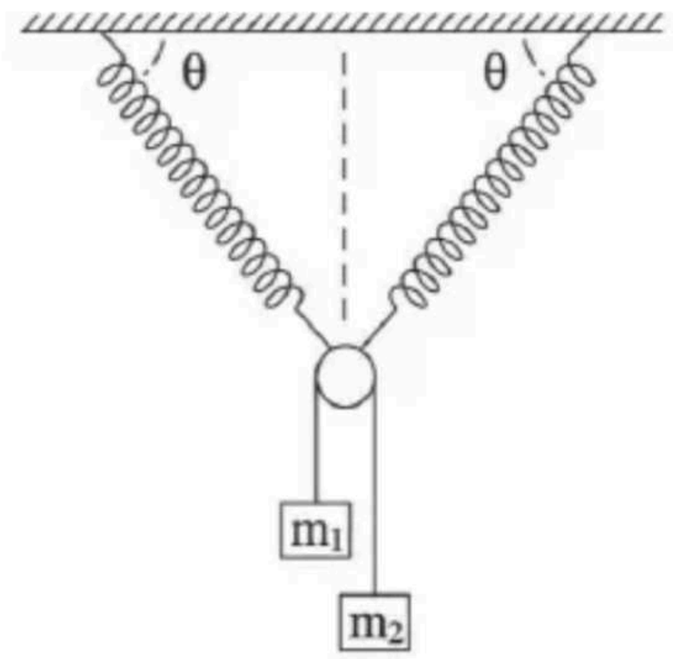
(3) 1.7 g cm^{-3} (4) 1.695 g cm^{-3}

94. Two stones are projected at the same speed but making different angles with the horizontal. Their horizontal ranges are equal. The angle of projection of one is $\pi/3$, and the maximum height reached by it is 102 m. Then the maximum height reached by the other in metres is

(1) 102 (2) 132

(3) 264 (4) 34

95. Two identical ideal springs of spring constant 1000 N/m are connected by a pulley as shown in the figure, and this arrangement is established in the vertical plane. The pulley is ideal and the string passing over the pulley is massless. When the pulley is at equilibrium, θ is 60° and the masses m_1 and m_2 are 2 kg and 3 kg, respectively. When θ is 60°, then the elongation in each spring is



(1) $(1.6)\sqrt{3}$ cm (2) $(4.8)\sqrt{3}$ cm

(3) 4.8 cm (4) $3.2/\sqrt{3}$ cm

96. The rear side of a truck is open, and a box of mass 20 kg is placed on the truck 4 m away from the open end, $\mu = 0.15$ and $g = 10\,m/s^2$. The truck starts from rest with an acceleration of $2\,m/s^2$ on a straight road. The box will fall off the truck when it is at a distance from the starting point equal to

(1) 4 m (2) 8 m (3) 16 m (4) 32 m

97. Read each statement below carefully.

Statement I : In an elastic collision of two bodies, the momentum and energy of each body is conserved.

Statement II : The total energy of a system is always conserved, no matter what internal and external forces on the body are present.

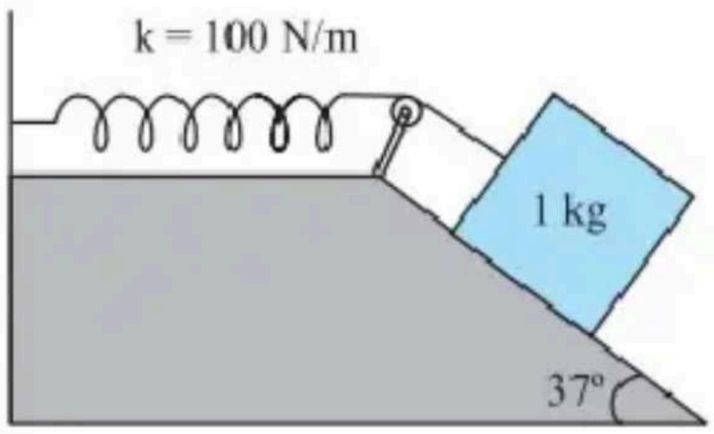
Statement III : Work done in the motion of a body over a closed loop is zero for every force in nature.

Statement IV : In an inelastic collision, the final kinetic energy can be less than the initial kinetic energy of the system.

In light of the above statements, choose the correct answer from the options given below.

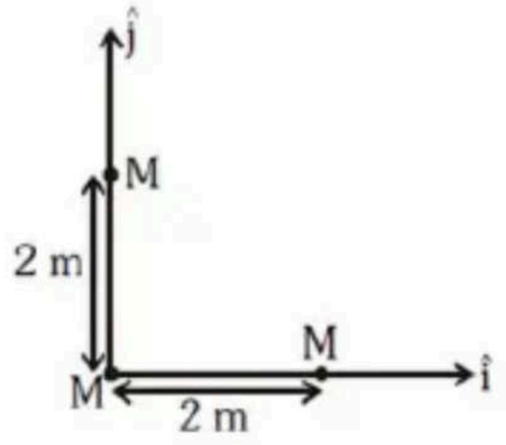
(1) II and IV are true
(2) II, III, and IV are true
(3) I, III, and IV are true
(4) Only IV is true

98. A 1-kg block situated on a rough incline is connected to a spring of spring constant 100 N/m as shown in the figure below. The block is released from rest with the spring in the unstretched position. The block moves 10 cm down the incline before coming to rest. The coefficient of friction between the block and the incline is nearly (Assume that the spring has negligible mass and the pulley is frictionless)



(1) 0.125 (2) 0.925
(3) 0.525 (4) 0.0125

99. Three identical spheres, each of mass M, are placed at the corners of a right-angle triangle with mutually perpendicular sides equal to 2 m (see figure). Taking the point of intersection of the two mutually perpendicular sides as the origin, find the position vector of the centre of mass (in m).



(1) $2\left(\hat{i}+\hat{j}\right)$
(2) $\left(\hat{i}+\hat{j}\right)$
(3) $\frac{2}{3}\left(\hat{i}+\hat{j}\right)$
(4) $\frac{4}{3}\left(\hat{i}+\hat{j}\right)$

100. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Machines such as the steam engine, the automobile engine, etc., which produce rotational motion, have a disc with a large moment of inertia, called a flywheel.

Reason (R) : Because of its large moment of inertia, the flywheel resists the sudden increase or decrease of the speed of the vehicle and prevents jerky motions, thereby ensuring a smooth ride.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(3) (A) is true, but (R) is false.
(4) (A) is false, but (R) is true.

101. A silver rod of length 1 m at 0°C is heated to 100°C. Its length is increased by 0.19 cm. The coefficient of cubical expansion of the silver rod is

(1) $5.7\times10^{-5}/^{\circ}C$
(2) $0.63\times10^{-5}/^{\circ}C$
(3) $1.9\times10^{-5}/^{\circ}C$
(4) $16.1\times10^{-5}/^{\circ}C$

102. A wall has two layers, A and B, made of different materials. The thickness of both layers is the same. The thermal conductivity of A and B are K_A and K_B such that $K_A = 3K_B$. The temperature across the wall is $20^\circ C$. In thermal equilibrium

(1) the temperature difference across A=15°C

(2) the temperature difference across A=5°C

(3) the temperature difference across A=10°C

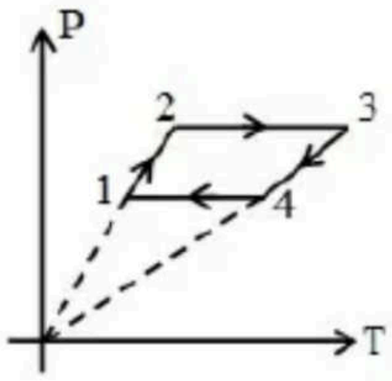
(4) the temperature difference across A=20°C

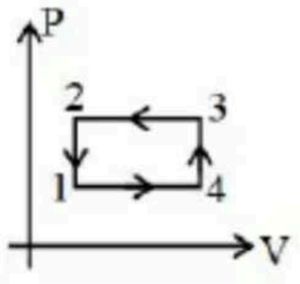
103. A mixture of 8 g of helium gas and 14 g of nitrogen gas is enclosed in a vessel of constant volume at 300 K. The quantity of heat absorbed by the mixture to double the r.m.s velocity of its molecules is (R = universal gas constant)

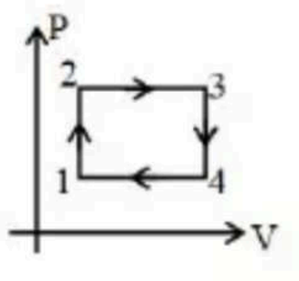
(1) $2725R$ (2) $3630R$

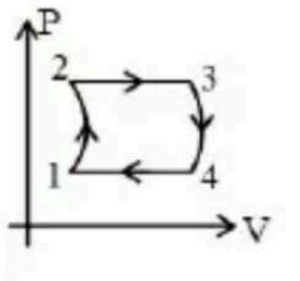
(3) $3825R$ (4) $5625R$

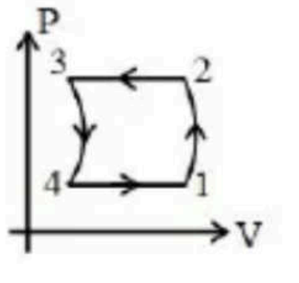
104. *P-T* graph of an ideal monoatomic gas is given as shown in the figure. The corresponding *P-V* diagram is



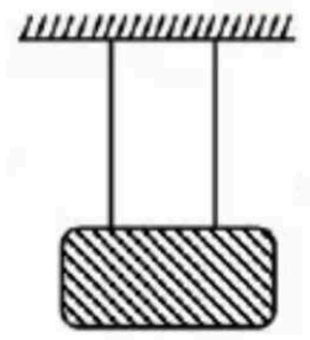
(1) 

(2) 

(3) 

(4) 

105. Two wires of equal length and equal cross-section are suspended as shown in the figure below. Their Young's moduli are Y_1 and Y_2 respectively. The equivalent Young's modulus will be



(1) $Y_1 + Y_2$ (2) $\frac{Y_1 + Y_2}{2}$

(3) $\frac{2Y_1Y_2}{Y_1 + Y_2}$ (4) $\sqrt{Y_1Y_2}$

106. A small hole with an area of cross-section $2\,mm^2$ is present near the bottom of a fully filled open tank of height 2 m. Taking $g = 10 m/s^2$, the rate of flow of water through the open hole would be nearly

(1) $8.9 \times 10^{-6} m^3/s$

(2) $12.6 \times 10^{-6} m^3/s$

(3) $2.23 \times 10^{-6} m^3/s$

(4) $6.4 \times 10^{-6} m^3/s$

107. Two small drops of mercury, each of radius R, coalesce to form a single large drop. The ratio of the total surface energies before and after the change is

(1) $1 : 2^{1/3}$ (2) $2^{1/3} : 1$

(3) 2 : 1 (4) 1 : 2

108. Two particles are oscillating along two close parallel straight lines side by side, with the same frequency and amplitudes. They pass each other, moving in opposite directions, when their displacement is half of the amplitude. The phase difference is

(1) $\frac{\pi}{6}$ (2) π

(3) $\frac{2\pi}{3}$ (4) zero

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109. The value of escape velocity on a certain planet is 2 km/s. Then the value of orbital speed for a satellite orbiting close to its surface is

(1) 12 km/s (2) 1 km/s

(3) $\sqrt{2}$ km/s (4) $2\sqrt{2}$ km/s

110. If the distance between the earth and the sun were half of its present value, then the number of days in a year would be

(1) 64.5 (2) 129

(3) 182.5 (4) 730

111. A string under a tension of 129.6 N produces 10 beats /sec when it is vibrated along with a tuning fork. When the tension in the string is increased to 160 N, it sounds in unison with the same tuning fork. Calculate the fundamental frequency of the tuning fork.

(1) 100 Hz (2) 50 Hz

(3) 90 Hz (4) 200 Hz

112. An electric dipole has a fixed dipole moment $\vec{p}$, which makes an angle θ with respect to the x-axis. When subjected to an electric field $\vec{E}_1 = E\hat{i}$, it experiences a torque $\vec{\tau}_1 = -\tau\hat{k}$. When subjected separately to another electric field $\vec{E}_2 = \sqrt{3}E\hat{j}$, it experiences a torque $\vec{\tau}_2 = -\vec{\tau}_1$. The angle θ is

(1) 30° (2) 45°

(3) 60° (4) 90°

113. A capacitor of $1\,\mu F$ withstands a maximum voltage of 6 kilovolts, while another capacitor of $2\,\mu F$ withstands a maximum voltage of 4 kilovolts. If the two capacitors are connected in series, the system will withstand a maximum voltage of

(1) 2 kV (2) 4 kV

(3) 6 kV (4) 9 kV

114. If n identical charged liquid drops of potential V merge into a big drop, then find the potential of the resulting drop.

(1) $n^{2/3}V$

(2) $n^{1/3}V$

(3) nV

(4) $V^{n/3}$

115. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1 kW. The voltage of the electric mains is 220 V. The minimum capacity of the main fuse of the building can be

(1) 8 A (2) 10 A

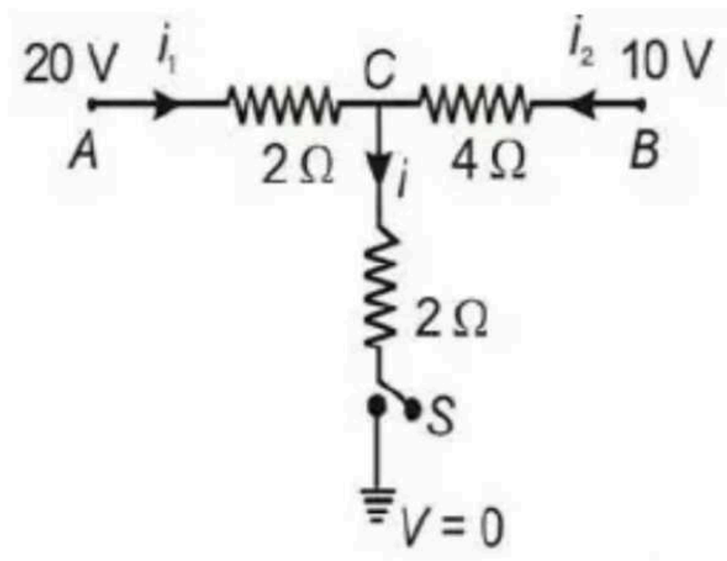
(3) 12 A (4) 11 A

116. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is $1\,k\Omega$. How much was the resistance on the left slot before interchanging the resistances?

(1) $990\,\Omega$ (2) $505\,\Omega$

(3) $550\,\Omega$ (4) $450\,\Omega$

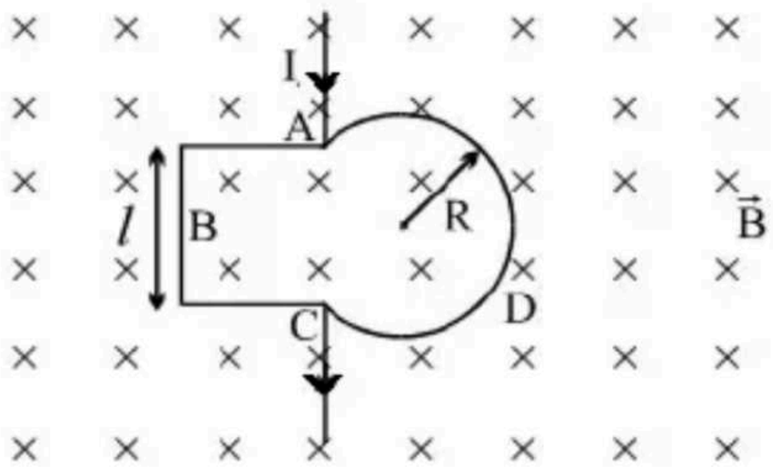
117. When the switch S, in the circuit shown below, is closed, then the value of current i through S will be



(1) 2 A (2) 5 A

(3) 4 A (4) 3 A

118. The figure given below shows a conducting loop ABCDA placed in a uniform magnetic field perpendicular to its plane. The part ABC is $(3/4)^{th}$ portion of the square of side length l. The part ADC is a circular arc of radius R. The points A and C are connected to a battery, which supplies a current I to the circuit. The magnetic force on the loop due to field B is



(1) zero

(2) BIl

(3) $2BIR$

(4) $\frac{BIlR}{l+R}$

119. An electron moving in a circular orbit of radius R makes n turns per second. The magnetic field at the centre has a magnitude of

(1) $\frac{2\mu_0 ne}{R}$ (2) $\frac{\mu_0 ne}{2R}$

(3) $\frac{\mu_0 ne}{\pi R}$ (4) zero

120. The magnetic susceptibility of a material at 200 K is -1.0×10^{-5}. Then its value at 400 K is _____ $\times10^{-5}$.

(1) -0.5 (2) -2.0

(3) -1.0 (4) 4.0

121. A conducting circular loop is placed in a uniform magnetic field 0.04 T with its plane perpendicular to the magnetic field. The radius of the loop starts shrinking at 2 mm/s. The induced emf in the loop when the radius is 2 cm is

(1) $1.6\pi\mu V$ (2) $3.2\pi\mu V$

(3) $4.8\pi\mu V$ (4) $0.8\pi\mu V$

122. In an LCR circuit, the voltage across the terminals of a resistance, inductance, and capacitance are 40 V, 30 V, and 60 V. Then, the voltage across the main source will be

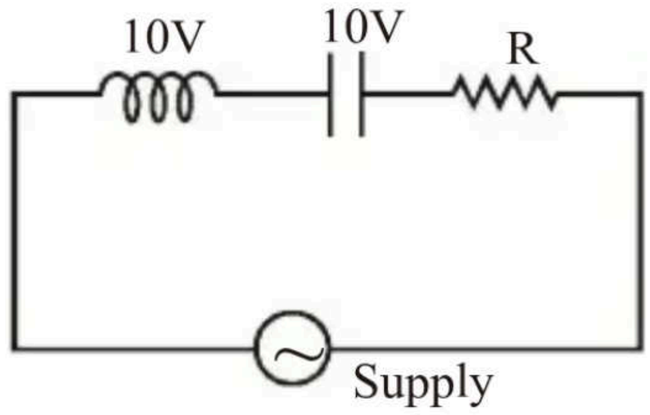
(1) 130 volt

(2) 100 volt

(3) 70 volt

(4) 50 volt

123. If the value of R is changed, then



(1) voltage across L remains the same

(2) voltage across C remains the same

(3) voltage across the LC combination remains the same

(4) voltage across the LC combination changes

124. The rms value of the electric field of light coming from the sun is 720 N/C. The average total energy density of the electromagnetic wave is

(1) $4.58\times10^{-6}\ J/m^3$

(2) $6.37\times10^{-9}\ J/m^3$

(3) $81.35\times10^{-12}\ J/m^3$

(4) $3.3\times10^{-3}\ J/m^3$

125. A plano-convex lens (μ = 1.5) has a radius of curvature 10 cm. It is silvered on its plane surface. Find the focal length after silvering.

(1) –10 cm (2) 20 cm

(3) –20 cm (4) 10 cm

126. A thin prism of angle 15° made of glass of refractive index μ_1=1.5 is combined with another prism of glass of refractive index μ_2 =1.75. The combination of the prisms produces dispersion without deviation. The angle of the second prism should be

(1) 5° (2) 7°

(3) 10° (4) 12°

127. For a concave mirror of focal length 20 cm, match the following.

	List I (Object Distance)		**List II** (Nature of Image)
A.	10 cm	I.	Magnified, inverted, and real
B.	30 cm	II.	Equal size, inverted, and real
C.	40 cm	III.	Smaller, inverted, and real
D.	50 cm	IV.	Magnified, erect and virtual

Choose the correct answer from the options given below.

(1) A-II B-III C-I D-IV

(2) A-IV B-I C-III D-II

(3) A-IV B-I C-II D-III

(4) A-I B-IV C-II D-III

128. Given below are two statements.

Statement I : Diffraction phenomena define the limits of ray optics.

Statement II : The speed of light in glass is independent of the colour of light.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

129. When a metallic surface is illuminated with the radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with the radiation of wavelength 2λ, the stopping potential is $V/4$. The threshold wavelength for the metallic surface is

(1) 4λ (2) 5λ

(3) $(5/2)\lambda$ (4) 3λ

130. A particle of mass 1 mg has the same wavelength as an electron moving with a velocity of 3×10^6 m/s. The velocity of the particle is (Mass of electron $=9.1\times10^{-31}$ kg)

(1) $2.7\times10^{-21}\,m/s$

(2) $2.7\times10^{-18}\,m/s$

(3) $9\times10^{-2}\,m/s$

(4) $3\times10^{-31}\,m/s$

131. The moment of momentum for an electron in the second orbit of hydrogen atom as per Bohr's model is

(1) π/h (2) 2πh

(3) $2h/\pi$ (4) h/π

132. If in a nuclear fusion reaction, mass defect is 0.3 %, then the energy released in the fusion of 1 kg of mass is

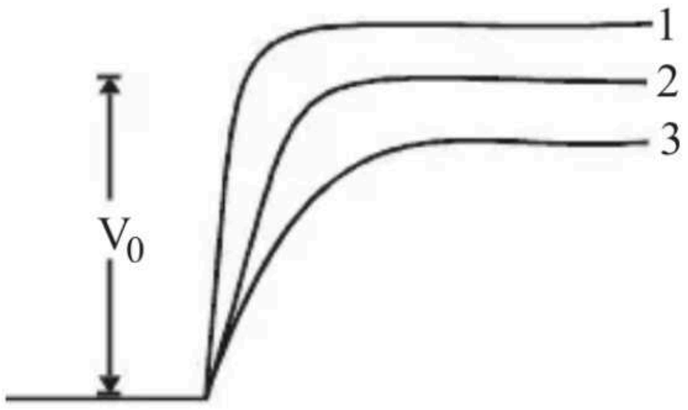
(1) 27×10^{10} J

(2) 27×10^{11} J

(3) 27×10^{12} J

(4) 27×10^{13} J

133. In the given figure, V_0 is the potential barrier across a p-n junction, when no battery is connected across the diode.



(1) Both curve 1 and curve 3 correspond to the forward bias of the junction.

(2) Curve 3 corresponds to the forward bias of the junction, and curve 1 corresponds to the reverse bias of the junction.

(3) Curve 1 corresponds to the forward bias, and curve 3 corresponds to the reverse bias of the junction.

(4) Both curve 3 and curve 1 correspond to the reverse bias of the junction.

134. In a p-n junction diode, change in temperature due to heating

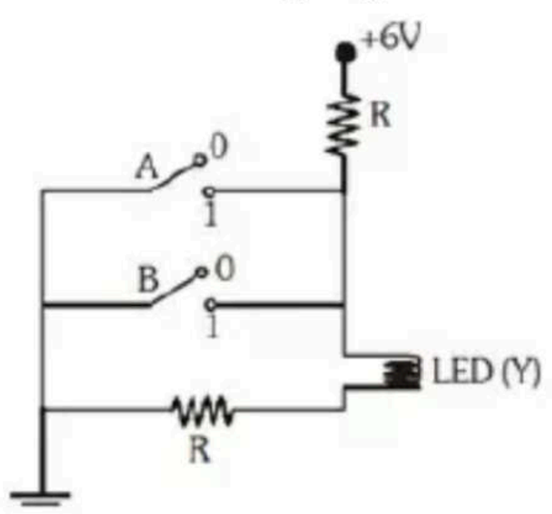
(1) affects only reverse resistance

(2) affects only forward resistance

(3) affects the overall V-I characteristics of the p-n junction

(4) does not affect the resistance of the p-n junction

135. The circuit diagram shown here corresponds to the logic gate

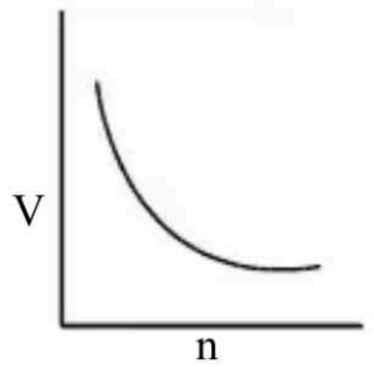


(1) NOR gate (2) AND gate

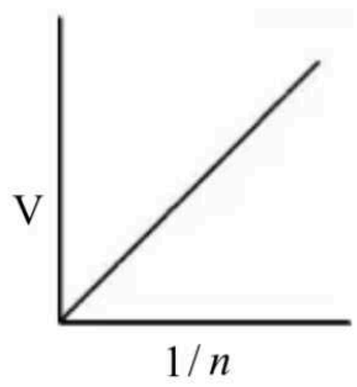
(3) OR gate (4) NAND gate

136. Select the graph that incorrectly represents how the velocity (V) of an electron in an orbit of a uni-electronic species is related with the principal quantum number (n) or atomic number (Z).

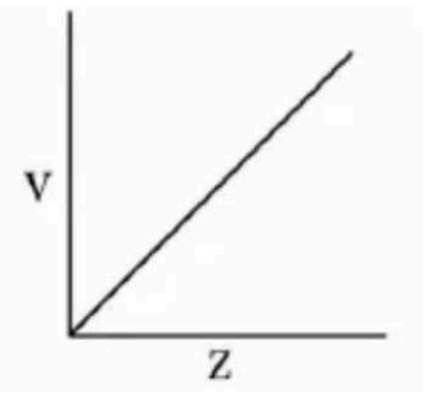
(1)



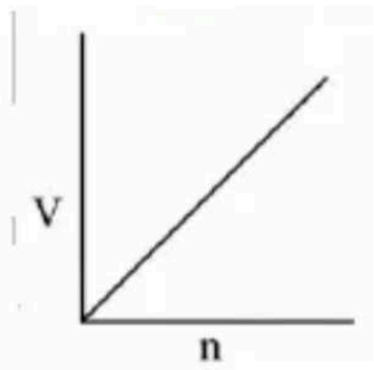
(2)



(3)



(4)



137. Select the correct statements from the list given below.

A. The number of 'm' [magnetic quantum number] values for a given 'n' are n^2.

B. The number of electrons spinning in the clockwise direction of a $3P_x$ orbital is 5.

C. The orbital with n = 3, l = 1, $m = -2$, $m_s = 0$ is designated as $3P_x$ atomic orbital.

D. The Eigen Wave function $\Psi(2,0,0)$ corresponds to 2s atomic orbital.

(1) Only A, B, D (2) Only A, B, C

(3) Only B, C (4) Only A, D

138. Select the correct statements from the list given below.

A. LiF is more ionic when compared to MgO.

B. HCN contains ionic, covalent, and dative bonds.

C. SO_2 is an electron-deficient molecule.

D. $\mu_{Total}\ of\ CH_3Cl > \mu_{Total}\ of\ CH_3F$

(1) Only A, C, D

(2) Only B, C, D

(3) Only C, D

(4) Only A, D

139. The maximum number of planar atoms are present in

(1) C_6H_6

(2) SF_6

(3) $CH_3 - C \equiv C - CH_3$

(4) PCl_5

140. The number of moles of $KMnO_4$ required to oxidise 0.5 mole of ferrous oxalate in an acidic medium is

(1) 0.3 (2) 0.5

(3) 1.6 (4) 0.75

141. The number of significant figures in 5.0201 is

(1) 3 (2) 2

(3) 5 (4) 4

142. A swimmer coming out of a swimming pool is covered with a film of water weighing 80 g. The amount of heat in KJ that is needed to evaporate this water is

$\left[\Delta H^0_{vap} = 40.79 kJ / mole\right]$

(1) 181 (2) 362

(3) 90.5 (4) 40.77

143. One mole of an ideal gas undergoes a change of state [2.0 atm, 3.0 L] to [2.0 atm, 7.0 L] with a change in internal energy of 30 L. atm. Then, the change in enthalpy (ΔH) of the process in KJ will be

(1) 38 (2) 3.8

(3) 7.6 (4) 42.7

144. When CO_2 gas bubbled through water for some time, the following equilibrium is established,

$CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$, with a K_C value of 3.8×10^{-7}. The pH of the system is found to be '6'. Then the ratio of concentration of HCO_3^- to CO_2 i.e., $\frac{\left[HCO_3^-\right]}{\left[CO_2\right]}$ will be

(1) 3.8×10^{-7}

(2) 3.8×10^{-13}

(3) 0.38

(4) 3.8×10^{-10}

145. Select the correct statements from the list given below.

A. pH of $\frac{M}{100}$ NaOH is 12 at 25°C.

B. pH of $\frac{M}{100}$ H_2SO_4 is 2 at 25°C.

C. pH of $\frac{M}{100}$ CH_3COOH is between 2 and 7.

D. pH of $\frac{M}{100}$ NaCl is 7.

(1) Only A, B, C

(2) Only A, C, D

(3) Only B, C

(4) Only C, D

146. $a = \left[AB, Ksp = 10^{-10} M^2 \right]$

$b = \left[AB_2, Ksp = 10^{-10} M^3 \right]$

$c = \left[A_3B_2, Ksp = 10^{-10} M^5 \right]$

$d = \left[AB_3, Ksp = 10^{-10} M^4 \right]$

of the sparingly soluble salts *a, b, c, d,* the solubility is maximum in water under identical conditions for

(1) *a* (2) *b*

(3) *c* (4) *d*

147. Select the correct statements.

A. SnF_4 and PbF_4 are ionic, whereas PbI_2 is covalent.

B. $SiCl_4 \xrightarrow[-4HCl]{H_2O(excess)} X \xrightarrow{\Delta} SiO_2 + H_2O$

C. The density of graphite is less than that of diamond.

D. In BF_3, boron atom is sp^2 hybridised.

(1) A, B, C, and D

(2) Only A, B, C

(3) Only B, C, D

(4) Only A, D

148. Two liquids, A and B, have vapour pressures in the ratio of $\left[P_A^0 : P_B^0 \right]$ 1: 3 at a certain temperature. Assume A and B form an ideal solution, and the mole fraction of A to B in the vapour phase is 4 : 3. Then the mole fraction of B in the solution state at the same temperature is

(1) $\frac{1}{5}$ (2) $\frac{2}{3}$

(3) $\frac{4}{5}$ (4) $\frac{1}{4}$

149. R.L.V. P is highest for an aqueous solution of

(1) 0.1m urea

(2) 0.1m NaCl

(3) 0.1m $MgCl_2$

(4) 0.1m $AlCl_3$

150. Limiting molar conductivities of Al^{+3} and SO_4^{2-} are $X\,\Omega^{-1}cm^2mole^{-1}$ and $Y\,\Omega^{-1}cm^2mole^{-1}$. Then the limiting molar conductivity of $Al_2(SO_4)_3$ will be

(1) X + Y

(2) 2X + 3Y

(3) $\frac{X+Y}{3}$

(4) $\frac{X+Y}{6}$

151. The time required for the complete decomposition of 4 moles of water using 4 amperes of current in seconds is

(1) 3.86×10^5

(2) 1.93×10^5

(3) 96500

(4) 48250

152. For a reaction $A+2B\rightarrow C$, the experimentally verified rate equation is Rate = K[A] [B]. If the concentration of 'B' is doubled keeping the concentration of 'A' constant, the value of the rate constant (K), at the same temperature, will

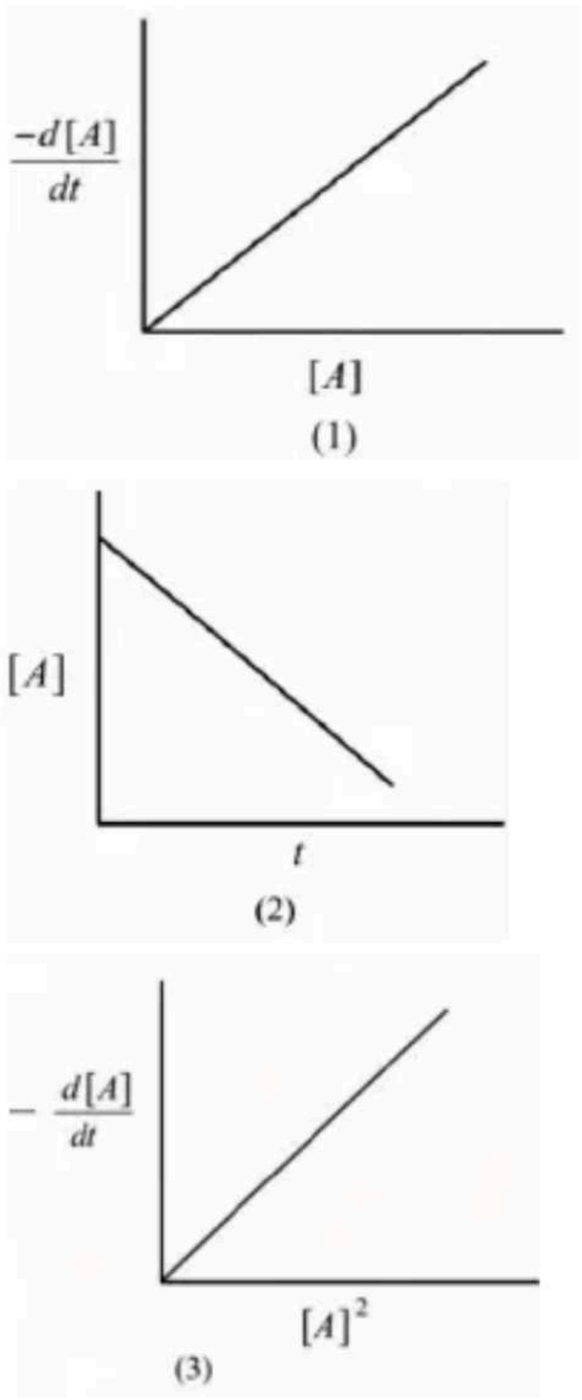
(1) remain the same

(2) get doubled

(3) get half

(4) get quadrupled

153. Consider the plots of the type, for the reaction $nA \rightarrow B + C$



Find which type of order of reaction each of these graphs represents?

(1) 0, 2, 1 (2) 0, 1, 2

(3) 1, 1, 2 (4) 1, 0, 2

154. Select the correct combination.

A. Possible oxidation states of Chlorine in its compounds = –1, +1, +3, +5, +7

B. Possible oxidation state of Sulphur its compounds = –2, +2, +4, +6

C. Possible oxidation state of Nitrogen its compounds = –3, +3, +5 only

D. Possible oxidation state of Oxygen its compounds = –1, –2, 0, +1, +2

(1) Only A and B

(2) Only A, B, D

(3) Only A, B, C

(4) Only B and C

155. Find the set, from the following, in which all are essential amino acids. [1 letter code of amino acid is given.]

(1) P, G, A (2) I, G, A

(3) Y, F, M (4) V, K, W

156. In the titration of $KMnO_4$ versus oxalic acid, in acidic medium which one acts as indicator?

(1) Methyl orange

(2) Phenolphthalein

(3) $(C_6H_5)_2NH$

(4) $KMnO_4$

157. A colourless salt gives brisk efflorescence with dil.H_2SO_4 and crimson red colour in the flame test. The salt is

(1) $SrSO_4$ (2) $BaCO_3$

(3) $SrCO_3$ (4) $CuCO_3$

158. The given complexes can exhibit which type of isomerism, respectively?

$[Cr(H_2O)_3(NH_3)_3]^{+3}$, $[Cr(en)_3]^{+3}$, and $[Cr(Gly)_3]$

(1) (Optical isomerism and Geometrical isomerism), (Optical isomerism) and (Optical isomerism and Geometrical isomerism)

(2) (Geometrical isomerism), (Optical isomerism) and (Geometrical isomerism) only

(3) (Geometrical isomerism), (Optical isomerism) and (Geometrical isomerism and Optical isomerism)

(4) (Optical isomerism and Geometrical isomerism), (Optical isomerism and Geometrical isomerism) and (Optical isomerism) only

159. Which of the following conversions is exothermic?

(1) $Na \rightarrow Na^+$ (2) $Li \rightarrow Li^-$

(3) $Ne \rightarrow Ne^-$ (4) $Mg \rightarrow Mg^+$

160. Which one of the following is the ***incorrect*** statement?

(1) The most EN element is A*l* among 13th group elements.

(2) The most EP elements are Cs among Li, Na, K, Rb, and Cs.

(3) IE of Xe is nearly equal to IE of the oxygen molecule.

(4) The magnitude of the ionisation enthalpy of Cs is less than that of C*l*.

161. The crystal field splitting of the Cr^{3+} ion in octahedral field changes for ligands I^-, H_2O, NH_3, CN^-. Then the correct increasing order of splitting is

(1) $I^- < H_2O < NH_3 < CN^-$

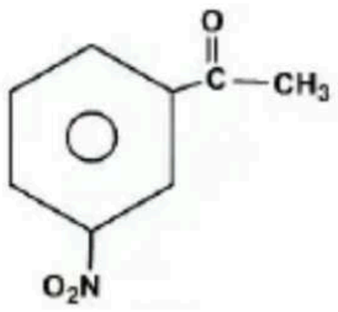
(2) $CN^- < I^- < H_2O < NH_3$

(3) $CN^- < NH_3 < H_2O < I^-$

(4) $NH_3 < H_2O < I^- < CN^-$

162. Given below are two statements.

Statement II : Benzene is treated with CH_3COCl / Lewis acid followed by Conc. HNO_3 + Conc. H_2SO_4 to give



Statement II : $-NO_2$ group is a meta-directing and deactivating group for EAS reaction.

In light of the above statements, choose the correct answer from the options given below.

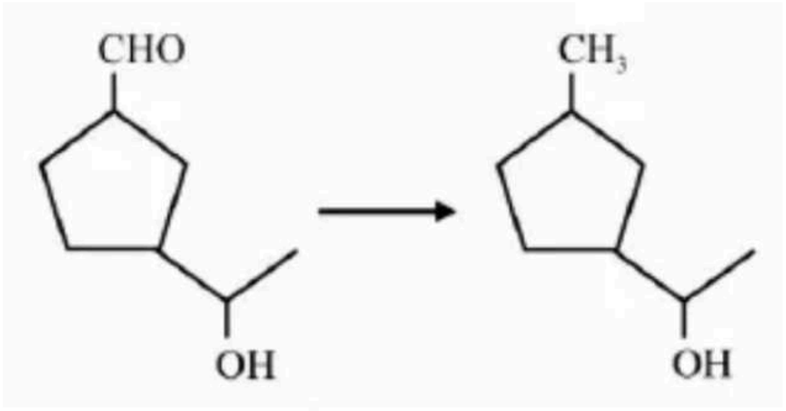
(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

163. The best reagent to carry out the following conversion is



(1) Red P + HI

(2) $LiAlH_4$

(3) Zn–Hg/HCl

(4) N_2H_4; KOH/Glycol/heat

164. Halogenation of an alkane is

(1) an electrophilic substitution reaction

(2) a nucleophilic addition reaction

(3) a free radical substitution reaction

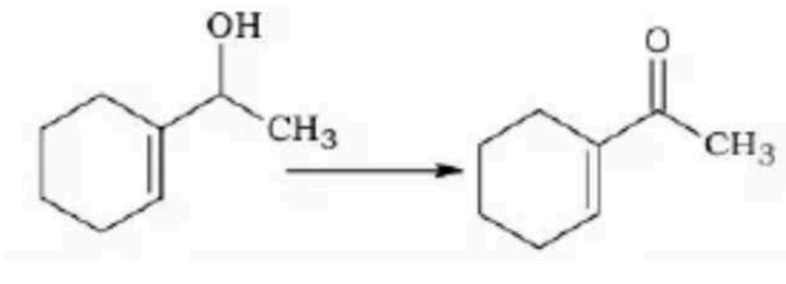
(4) a free radical addition reaction

165. In 3,3-dimethylhex -1- ene -4-yne, there are ____ sp^3, ___ sp^2 and ___ sp hybridised carbon atoms respectively.

(1) 4, 2, 2 (2) 3 ,3, 2

(3) 2, 4, 2 (4) 2, 2, 4

166. Among the following, the number of reagents that affect the given conversion is



A. PCC B. PDC

C. $KMnO_4$/Conc.H_2SO_4

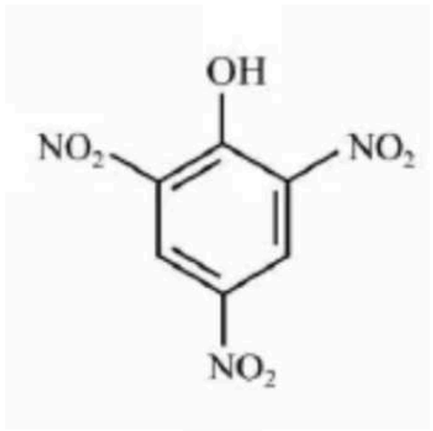
D. Br_2/NaOH

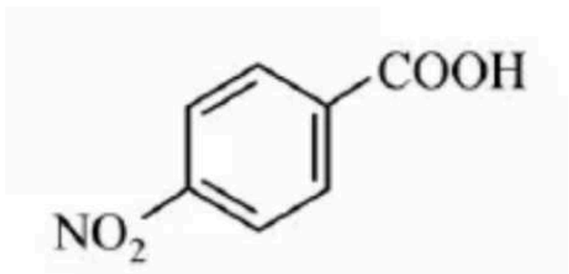
E. Alkaline $KMnO_4$

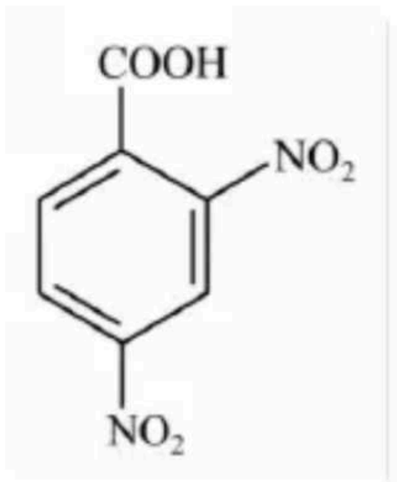
(1) four (2) two

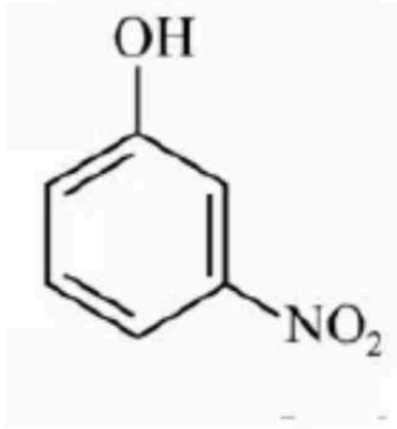
(3) five (4) three

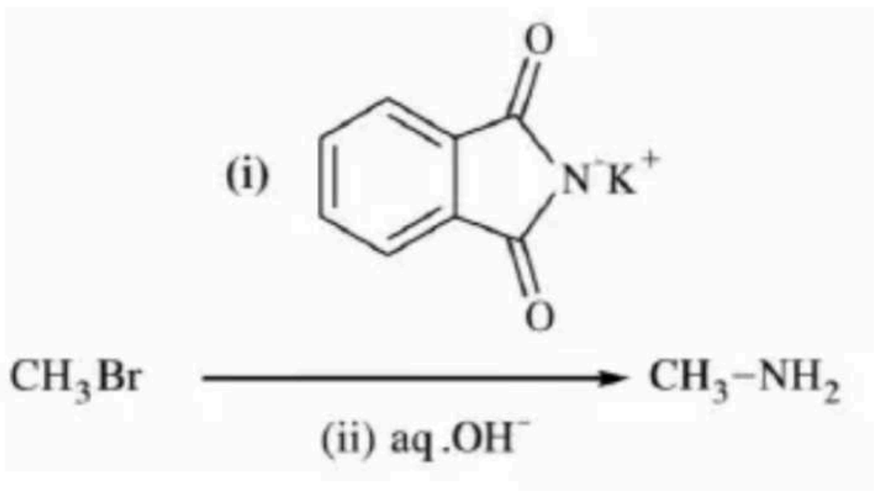
167. Which of the following compounds is least likely to give effervescence of CO_2 in the presence of aq. $NaHCO_3$?

(1) 

(2) 

(3) 

(4) 

168. 

This reaction is known as

(1) Hoffmann degradation

(2) Coupling reaction

(3) Carbylamine reaction

(4) Gabriel phthalimide synthesis

169. Given below are two statements.

Statement I : Hyperconjugation is not a permanent effect.

Statement II : Carbocation stability increases with an increase in the number of alpha (α) C–H bonds due to enhanced hyperconjugation.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

170. Allotrope of carbon used as filler in automobile tyres is

(1) diamond

(2) graphite

(3) fullerene

(4) carbon black

171. Formalin is

(1) 4% formaldehyde solution

(2) 40% formaldehyde solution

(3) 100% formaldehyde solution

(4) 87% formaldehyde solution

172. The Kjeldahl method is ***not*** applicable to

(1) aniline

(2) pyrrole

(3) benzylamine

(4) methanamine

173. Match List I with List II.

	List I		List II
A.	Carbocation	I.	Species that can supply a pair of electrons
B.	Methyl free radical	II.	Species that can receive a pair of electrons
C.	Nucleophile	III.	Sp^2 hybridised carbon with empty p-orbital
D.	Electrophile	IV.	Paramagnetic with one unpaired electron

Choose the correct answer from the options given below.

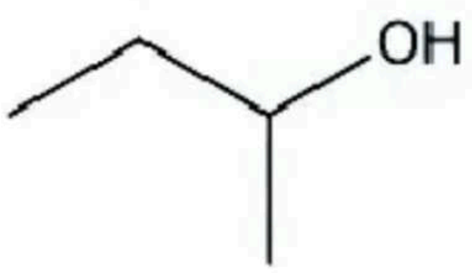
(1) A-IV B-II C-III D-I

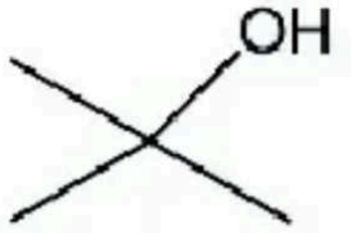
(2) A-II B-III C-I D-IV

(3) A-III B-IV C-II D-I

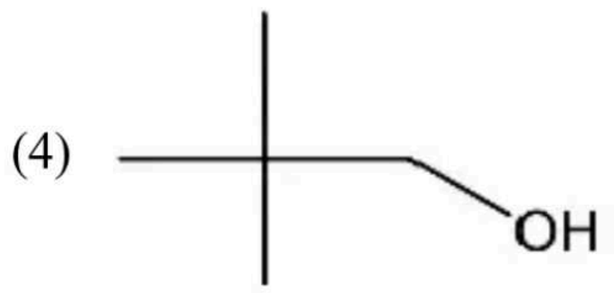
(4) A-III B-IV C-I D-II

174. Which of the following alcohols ***cannot*** be prepared from the hydration of an alkene?

(1) 

(2) 

(3) 

(4) 

175. Identify the correct major product.

$$CH_3 - C \equiv N \xrightarrow[(ii)\,HOH,\,H^+]{(i)\,SnCl_2 + HCl} \text{Product}$$

(1) $CH_3 - \overset{\overset{O}{\|}}{C} - OH$

(2) $CH_3 - \overset{\overset{O}{\|}}{C} - H$

(3) $CH_3 - CH_2 - OH$

(4) $CH_3 - CH_2 - NH_2$

176. The maximum oxidation state possible for U and Np, respectively, are

(1) 5 and 7

(2) 6 and 6

(3) 7 and 7

(4) 6 and 7

177. Which of the following statements is correct for a 3d-transition element?

(1) All the metals except Sc form 'MO' oxide.

(2) All the metals except Zn form 'MO' oxide.

(3) All the metals except Zn and Sc form ‘MO' oxide.

(4) All the metals except Mn form ‘MO' oxide.

178. The ***incorrect*** statements regarding geometrical isomerism are

A. Propene shows geometrical isomerism.

B. The trans isomer has identical atoms/groups on the diagonally opposite sides of the double bond.

C. Cis-but-2-ene has a higher dipole moment than trans but-2-ene.

D. 2-methylbut-2-ene shows two geometrical isomers.

E. The trans isomer has a lower melting point than the cis isomer.

Choose the correct answer from the options given below

(1) A, D, and E only

(2) C, D, and E only

(3) B and C only

(4) A and E only

179. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Alkyl fluorides are more reactive than alkyl iodides towards nucleophilic substitution reactions.

Reason (R) : The leaving group ability of iodide is more than that of fluoride.

In light of the above statements, choose the correct answer from the options given below.

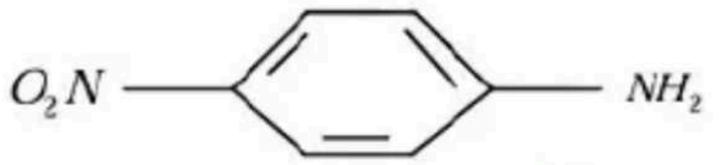
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

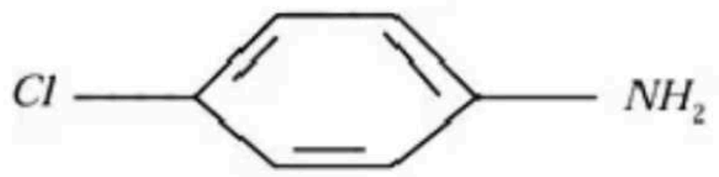
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

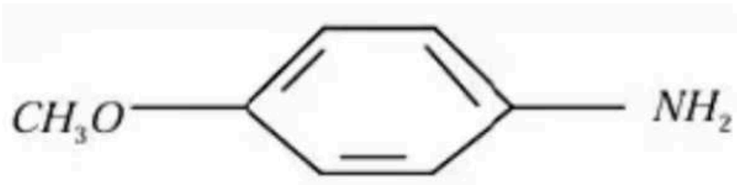
(3) (A) is true, but (R) is false.

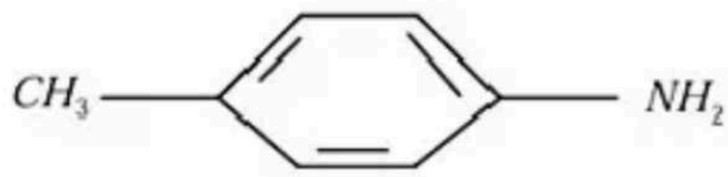
(4) (A) is false, but (R) is true.

180. Which of the following arylamines undergoes diazotisation with great difficulty?

(1) 

(2) 

(3) 

(4) 

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